

Hands-on: Accelerating k-NN with Rcpp

Overview

In this 30 min session you will:

1. Download and inspect the C++ implementation for k-NN.
 2. Compile and load the Rcpp code.
 3. Run benchmarks comparing pure-R vs Rcpp predictions.
 4. Analyze how sample size and dimensionality affect performance.
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Setup

Download the C++ and helper scripts into your working directory:

```
1 curl -O https://raw.githubusercontent.com/mmadoliat/KNNreg/refs/heads/main/R/knn_pred.R
2 curl -O https://raw.githubusercontent.com/mmadoliat/KNNreg/refs/heads/main/src/knn_pred.cpp
3 curl -O https://raw.githubusercontent.com/mmadoliat/KNNreg/refs/heads/main/inst/examples/KNNreg.R
```

Open the files in your editor to review the code:

- **knn_pred.R** contains the `knn_pred()` function (R).
 - **knn_pred.cpp** contains the `knn_pred_cpp()` function (Rcpp).
 - **KNNreg-Rcpp.R** sources both R and C++ implementations and runs `microbenchmark()`.
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1. Compile the C++ code

In an R console or RStudio, run:

```
1 Rcpp::sourceCpp("knn_pred.cpp")
```

If successful, you should see `knn_pred_cpp` available:

```
1 ls("package:base") # confirm knn_pred_cpp is loaded
2 # [1] "knn_pred_cpp"
```

2. Inspect the runner script

Open **KNNreg-Rcpp.R**, which contains:

```
1 library(microbenchmark)
2
3 # compile Rcpp code once at startup
4 Rcpp::sourceCpp("knn_pred.cpp")
5
6 # load R functions
7 source("knn_pred.R")
8
9 set.seed(42)
10 n <- 1000; p <- 5; m <- 200; k <- 10
11 Xtrain <- matrix(rnorm(n*p), n, p)
12 Ytrain <- rnorm(n)
13 Xtest  <- matrix(rnorm(m*p), m, p)
14
15 # warm up & check correctness
16 pred_R <- knn_pred(Xtrain, Ytrain, Xtest, k, method="R")
17 pred_C <- knn_pred(Xtrain, Ytrain, Xtest, k, method="cpp")
18 stopifnot(max(abs(pred_R$preds - pred_C$preds)) < 1e-12)
19
20 # benchmark
21 mb <- microbenchmark(
22   pure_R = knn_pred(Xtrain, Ytrain, Xtest, k, method="R"),
23   Rcpp   = knn_pred(Xtrain, Ytrain, Xtest, k, method="cpp"),
24   times  = 20
```

```
25 )  
26 print(mb)
```

Try running this script:

```
1 source("inst/examples/KNNreg-Rcpp.R")
```

3. Vary parameters

Modify **KNNreg-Rcpp.R** or re-run interactively to examine different settings:

- Increase **n** (training size) from 1000 to 5000 or 10000.
- Increase **p** (dimensions) from 5 to 20 or 50.
- Observe how the Rcpp version scales relative to pure-R.

Focus on how the Rcpp implementation stays much faster as complexity grows.

4. Discussion

- Where does Rcpp help most?
- Are there settings where pure R is sufficient?
- How might you further optimize (e.g., using STL `partial_sort`)?

Next steps

- Try integrating this into a Shiny app.
- Explore parallel Rcpp implementations (OpenMP).
- Consider other statistical routines with nested loops.